Department of Artificial Intelligence, SVNIT,SURAT
Subject- Machine Learning(AI301)

LAB ASSIGNMENT-6

1.

Objectives: Handling Class Imbalance and Hypothesis Testing

Dataset: Use the **Credit Card Fraud Detection dataset** (available on Kaggle).

- 284,807 transactions (rows).
- "Class" column = 0 (non-fraud) or 1 (fraud).
- Fraud cases are ~0.17% → strong class imbalance.

Part A : Perform EDA on dataset

Part B : **Exploring Class Imbalance**

1. Plot class distribution (pie chart / bar plot).
2. Discuss why class imbalance is problematic for machine learning models.

Part C: Under-sampling and Over-sampling

1. Apply **Random Under-Sampling (RUS)**: balance classes by downsampling majority class.
2. Apply **Random Over-Sampling (ROS)** or **SMOTE** (Synthetic Minority Over-sampling Technique).

Part D: Discussion & Insights

1. Which method (under/over-sampling) worked best and why?
2. Which metrics (precision, recall, F1) are most useful for fraud detection?
3. How does hypothesis testing support feature selection in imbalanced datasets?

2.

You are given a dataset of SMS messages labeled as spam or ham.

1. Preprocess the dataset by:
   - o Converting all text to lowercase.
   - o Removing stop words and punctuation.
   - o Converting the text into numerical features using CountVectorizer or TfidfVectorizer.
2. Train a Bernoulli Naive Bayes classifier on the processed data.
3. Compare its performance with a Multinomial Naive Bayes classifier using accuracy, precision, recall, and F1-score.
4. Based on your results:
   - o Which model performs better for SMS spam detection?
   - o Explain why Bernoulli NB may be more suitable for binary features, while Multinomial NB is generally better for word count features.